Extracellular matrix control over physiological and pathological wound: insights from a new wound-on-a-chip model

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Introduction

Normal wound healing is a tightly regulated process involving coordinated inflammation, proliferation, and remodeling [1]. In keloids, persistent fibroblast activation and excessive extracellular matrix (ECM) deposition lead to disorganized collagen networks [2]. In vitro wound models rarely control ECM architecture across the wound interface, and wound-on-a-chip devices mainly address inflammation or re-epithelialization [3]. We developed a wound healing-on-a-chip platform to study how fibroblast phenotype and matrix density shape fibroblast–ECM interactions in normal and keloid healing.

Methods

Human dermal (HDFs) and keloid fibroblasts (KDFs) are embedded in 2.5 mg/ml collagen and loaded into the lateral channels of a 3D microfluidic device, flanking a central acellular chamber with 3 or 6 mg/ml matrices. On day 7, recruitment (cells invading the central chamber) and maximum invasion are measured by phase contrast live imaging. Pore and fiber diameter of acellular gels and collagen coherency, fiber thickness and interfiber spacing of cellular matrices are quantified from multiphoton second harmonic generation (SHG) imaging.

Results

Higher collagen concentration alters matrix architecture, not only stiffness. In acellular gels, fiber diameter increases significantly at 6 mg/ml (1.70 compared to 1.58 and 1.56 µm) while pore diameter decreases non-significantly (7.1 vs 9.3 µm).

Cell type is the main determinant of recruitment. KDFs recruit fewer cells than HDFs, significantly at 6 mg/ml (Figure 1). Density of the matrices has no significant effect, and invasion distance does not differ between cell types or densities.

Despite low recruitment, KDFs appears to remodel the remote matrix. At 6 mg/ml, all three KDF devices show higher central-channel alignment (coherency fold change vs acellular: 0.65 lateral channel vs 1.48 central channel) and interfiber spacing (0.97 vs 1.28), with thinner fibers than in acellular gels (0.84–0.93), whereas HDF responses are heterogeneous (Figure 2). These trends, consistent with KDFs pulling on the matrix, are not significant.

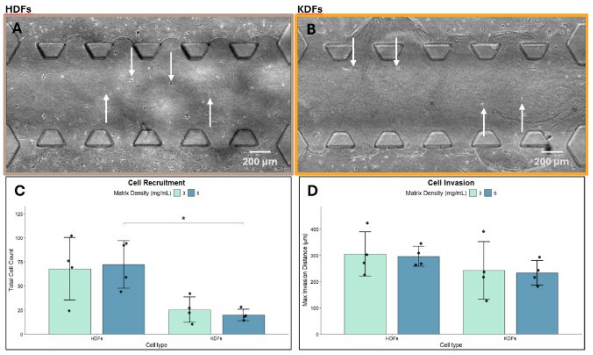


*Figure 1: Day 7 phase-contrast images of the central chamber (6 mg/ml) with (A) HDFs and (B) KDFs; arrows: invading fibroblasts; scale bars: 200 µm. (C) Recruitment and (D) maximum invasion distance (mean ± SD, n = 4 devices; two-way ANOVA + Tukey; *p < 0.05).*

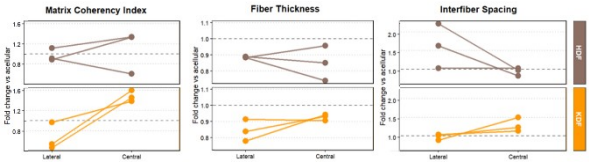


Figure 2: (A) Coherency, (B) fiber thickness and (C) interfiber spacing in lateral and central channels of 6 mg/ml devices at day 7, as fold change vs acellular devices (n = 3 devices).

Conclusions

KDFs show markedly lower recruitment than HDFs regardless of matrix density, but appear to remodel the remote matrix, increasing collagen alignment and fiber spacing. Additional biological replicates are being analyzed. This wound-on-a-chip provides a 3D tool to study fibroblast–ECM interactions in pathological healing.

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References

1. Lacina, L. et al, *Front Immunol*, *15*, 2024
2. Limandjaja GC. et al, Front Cell Dev Biol., 8:360, 2020.
3. Biglari S. et al, Adv Healthc Mater, 8:e1801307, 2019.